Extended reading — Week 03: Monte Carlo Methods

October 20, 2026

Read with a question in mind

Start with the week's teaching guide and its small worked example. Then use these readings to connect the algorithm to its assumptions and evidence. The explanations and investigations below are course commentary, not quotations. Read the source before reusing a theorem or a reported result. For each paper, record the objective, observation model, claimed result, conditions and one condition your local experiment does not reproduce.

1. Robbins and Monro (1951): A Stochastic Approximation Method

Primary source or book record: https://doi.org/10.1214/aoms/1177729586.

What to learn. Studies noisy iterative root finding, motivating updates that average observations over time. Conditions such as $\sum_t \alpha_t = \infty$ and $\sum_t \alpha_t^2 < \infty$ recur in stochastic approximation.

What the result requires. Those step-size conditions alone are not a complete convergence theorem: the drift, noise and stability assumptions matter. Constant step sizes can track change but do not satisfy the square-summability condition.

Notebook investigation. Compare incremental averaging with the explicit sample mean. In `exercise_1_first_visit_returns`, identify which episodes contribute one observation for a state.

2. Singh and Sutton (1996): Reinforcement Learning with Replacing Eligibility Traces

Primary source or book record: https://doi.org/10.1007/BF00114726.

What to learn. Connects visitation conventions and eligibility traces. Read the Monte Carlo discussion alongside the distinction between counting one return per episode and pooling all visits.

What the result requires. First-visit returns can be averaged as independent episode-level observations under the usual sampling model. Every-visit estimates can be consistent while biased at finite sample size because visits and the denominator are dependent. Neither method has universally lower variance.

Notebook investigation. Use `exercise_2_every_visit_mc`. Create one episode that repeats a state and another that does not. Compare pooling visits with averaging one return per episode; state the denominator of each estimator.

3. Precup, Sutton and Singh (2000): Eligibility Traces for Off-Policy Policy Evaluation

Primary source or book record: https://web.eecs.umich.edu/~baveja/Papers/OffPolicy.pdf.

What to learn. Uses importance sampling to connect behaviour-policy data with a target policy in temporal-difference and trace methods. Per-decision weighting avoids applying future action ratios to an earlier reward.

What the result requires. Every target-supported decision must have behaviour support. Shorter ratio products can help, but do not establish a universal variance ordering or recover rewards for never-selected actions.

Notebook investigation. For a two-step trajectory, derive $\rho_0 r_1 + \gamma \rho_0 \rho_1 r_2$. Compare with multiplying the whole return by $\rho_0 \rho_1$ and explain why the first reward does not require the second action ratio.

4. Jiang and Li (2016); Thomas and Brunskill (2016): Sequential Doubly Robust Evaluation

Primary source or book record: https://proceedings.mlr.press/v48/jiang16.html.

What to learn. Jiang and Li extend doubly robust evaluation to sequential decisions. Thomas and Brunskill develop data-efficient estimators including weighted DR and MAGIC; their paper is linked below. These are distinct contributions.

What the result requires. Double robustness does not mean lower variance than every alternative for every reward model. Weighted estimators can introduce finite-sample bias. The Thomas–Brunskill paper is not the original high-confidence off-policy evaluation paper.

Notebook investigation. Use `exercise_4_ordinary_is` and `exercise_5_weighted_is` before adding a model. Track overlap, weight tails and repeated-sample error. Additional source: https://proceedings.mlr.press/v48/thomasa16.html.

5. Joachims, Swaminathan and de Rijke (2018): Deep Learning with Logged Bandit Feedback

Primary source or book record: https://www.cs.cornell.edu/~tj/publications/joachims_etal_18a.pdf.

What to learn. Develops BanditNet for learning a deep policy from logged bandit feedback, addressing optimisation difficulties of counterfactual objectives.

What the result requires. A training objective evaluated on the data used to select a policy is not an independent estimate of that policy value. Importance weighting corrects an observation mechanism under support and correct propensities; it is not intrinsically a source of bias.

Notebook investigation. In `exercise_7_offpolicy_recruitment`, separate policy selection and held-out evaluation. Keep clipping choices, reward transformations and denominators explicit.

Synthesis: evidence you should be able to show

1. Reproduce one numerical calculation before running a long experiment.
2. Name the precise objective, sampling distribution and terminal convention.
3. Compare the local implementation with one source: list a shared mechanism and one simplification that prevents calling it a full reproduction.
4. Separate an algebraic identity, an assumption-dependent theorem and an empirical result. A plot alone does not prove a theorem.
5. Save the notebook with seed, parameters, axes, denominators and a short explanation of an unexpected result. Keep evaluation data separate from tuning.

Return to the teaching guide's notebook investigation for the local function names, expected checks and interpretation questions. Source code and notebooks are available together in the course repository; the web reader is the entry point and the downloaded notebook is the execution workspace.